

Trigonometric identities



1 $\frac{\sin x}{\cos x}$

2 a $\frac{1}{2}$

b $\frac{a}{c}$

3 a $\frac{8}{17}$

b $-\frac{12}{13}$

4 a $\frac{4}{3}$

b $\frac{2}{\sqrt{5}}$

5 $\csc \theta = \frac{1}{\sin \theta}$

6 a $\cot \theta$

b $\sin^2(\theta)$

c $\sec^2(x)$

d $\cot \theta$

e $\tan \theta$

f 1

g $\sin \theta$

h 1

i $\sin y$

j $\tan \theta - 1$

k $\sin^2(\theta)$

l $\csc \theta$

m $\sec \theta$

7 a 2

b $\frac{17}{31}$

c $\frac{11}{6}$

d $\pm \frac{1}{\sqrt{2}}$

8 $\frac{9}{2}$

9 $\frac{7}{4}$

10 $\frac{1}{11}$

11 $\frac{3}{8}$

12  a $\frac{\sqrt{3}}{3}$ b $-\frac{\sqrt{6}}{4}$

13 Impossible. If $\tan \theta = 3.5$ then $\cot \theta = \frac{1}{3.5} \approx 0.29 \neq -3.5$.

Pythagorean identities

14 a Yes b Yes c No d No

15 Impossible

16 a $\sin \theta$ b $-\sin^2(\theta)$ c $1 - 2 \cos \theta \sin \theta$ d $-\cos^3(\theta)$
e $10 - 6 \cos x$ f 1 g $-\tan \theta$ h 1
i $\sin^2(\theta)$ j $-\cot \theta \cos \theta$ k $-\tan^2(x)$ l $\tan \theta - \cot \theta$

17 a $\tan^2(\theta)$ b $\cos^2(\theta)$ c $\cos^2(x)$ d $\csc^2(\theta)$
e $\sec^2(\theta)$ f $\tan^4(x)$ g $\tan^2(\theta)$

18 a $2\csc^2(\theta)$ b $\cot^2(x)$ c $\tan^2(x)$ d $\sec \theta \csc \theta$
e $\sec^2(x)$ f $\cot^2(x)$ g $2\csc \theta$ h 1
i 1 j $2 \cos \theta \csc^2(\theta)$ k $-\cot x$ l $\cot \theta \cos \theta$

19 $a \sec \theta$

20 $\cot x = \pm \sqrt{\csc^2(x) - 1}$

21 a $\sin x = \frac{5}{13}$ b $\tan x = \frac{5}{12}$

22 $\dots$ 1

